Impact of Indian science, humanities and technology on transformative development and SDG based goals and action-plans

Role of Humanities | Science & Technology | Action Plans

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What is Transformative Development?

Development that transforms systems and institutions - not just statistics.

1

Structural Change

2

Inclusive Growth

3

Environmental Sustainability

4

Institutional Change

5

Human Development

India's SDG Performance

99th

Global SDG Index Rank

out of 167 countries

71/100

India's SDG Score

4th Edition SDG India Index

67

International Score

Global SDG Index Score

Key Challenges Remaining:

SDG 2

Zero Hunger

GHI Score: 25.8 (2025)

Classified as 'Serious' hunger level

SDG 5

Gender Equality

GII: 0.437 (2022)

Per UNDP Human Development Report

SDG 10

Reduced Inequalities

Gini Index: 25.5 (2022)

Per World Bank data

Role of Indian Humanities in Transformative Development

While science provides tools, humanities determine how they are used — and for whom.

01 Ethical Governance & Public Policy

SDGs: SDG 16, 10, 5, 1, 17
Thinkers like Ambedkar, Gandhi, and Amartya Sen shaped inclusive governance. Political philosophy drives institutional accountability in bodies like NITI Aayog.

02 Understanding Social Inequality & Poverty

SDGs: SDG 1, 10, 2, 8
Sociology & anthropology reveal structural causes of poverty. Informed schemes like MGNREGA, reservation policies, and tribal rights legislation.

03 Promoting Gender Equality & Social Justice

SDGs: SDG 5, 4, 3, 8
Gender studies exposed patriarchal norms, shaping Beti Bachao Beti Padhao, Sukanya Samriddhi Yojana, and PM Ujjwala Yojana.

04 Preserving Cultural Heritage & Communities

SDGs: SDG 11, 8, 4
History, archaeology & cultural studies support sustainable tourism, creative industries, and local livelihoods through institutions like ASI and IGNCA.

Impact of Science & Technology on SDGs

Education & Digital Learning

SDG 4

Digital platforms, databases & online delivery extended quality education to remote regions.

Healthcare & Medical Technology

SDG 3

Ayushman Bharat, digital health records & National Health Mission expanded affordable care.

Agriculture & Food Security

SDG 2

PM-KISAN, climate-resilient crops & soil health tech improved farmer resilience & productivity.

Clean Energy & Environment

SDG 7, 13

Green Hydrogen Mission, solar & wind energy drive fossil fuel transition & reduce emissions.

Digital Governance

SDG 9, 16

Aadhaar, UPI & Digital Public Infrastructure expanded financial inclusion & transparent governance.

Action Plan: Convergence of Technology & Humanities

1

SDG Localisation through Data + Community Knowledge

Pair data analytics (SDG India Index) with ethnographic research to capture barriers like caste, language, and historical marginalisation at sub-national level.

2

Ethical Governance of Digital Public Infrastructure

Embed data ethics, digital rights, and privacy law into governance of Aadhaar, UPI, and AI-assisted welfare systems to prevent algorithmic bias.

3

Behavioural Science + Medical Technology

Combine telemedicine & diagnostic AI with anthropology and psychology to address health-seeking behaviour, vaccination hesitancy, and mental health stigma.

4

Climate Science + Environmental Humanities

Alongside Green Hydrogen Mission & PM-KUSUM, integrate energy justice, land rights, and participatory dialogue for communities affected by clean energy transitions.

5

Viksit Bharat@2047: Innovation + Liberal Tradition

Build a $30 trillion economy with concurrent investment in humanities — historians, philosophers, and social scientists who can interrogate power and articulate rights.

Conclusion

Uneven but Real Progress

India's SDG Index score of 71 and expanding digital infrastructure show institutional will, but hunger, gender equality, and economic inequality remain persistent.

Technology as Instrument

Digital learning, Ayushman Bharat, Green Hydrogen, and Aadhaar/UPI have reshaped governance, but technology alone is insufficient for transformative change.

Values Must Guide Innovation

The SDGs are ultimately a humanistic commitment. Data must inform policy, but values must guide it. India's path to Viksit Bharat requires both technological ambition and humanistic responsibility.

"The SDGs are not a technocratic checklist - they are a moral and political commitment to the dignity of every human being."

Thank You!